# PHYSICS I
# WEEK
# 13 & 14

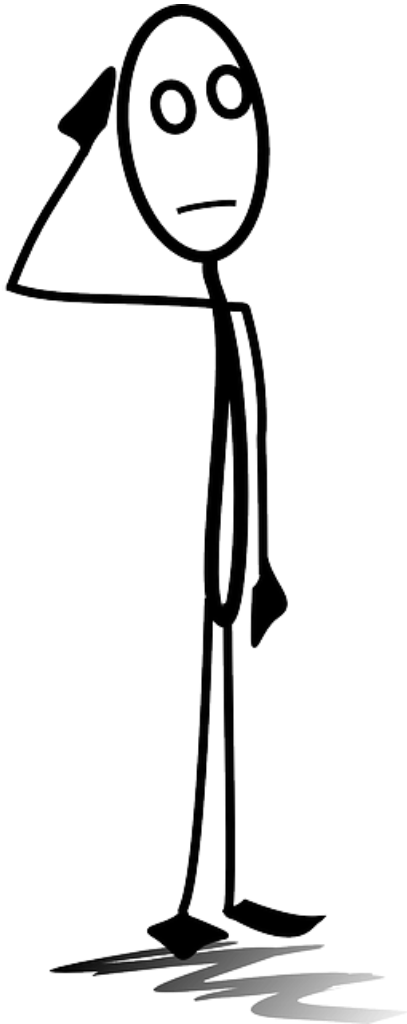


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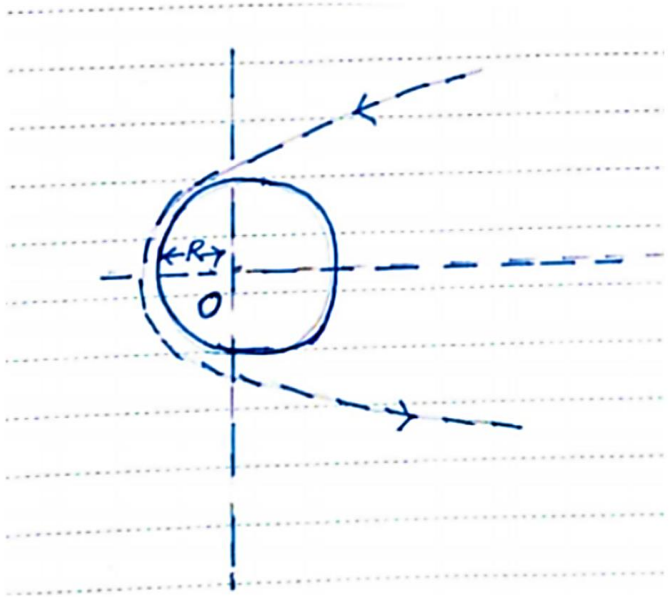
# WEEK 13 & 14: OUTLINE

- Tutorial 8
- Revision: Vector Calculus – Gauss divergence theorem & Stokes' theorem (Schey Chapter II & III)
- Fluid Dynamics: Taylor 16.12
- Tutorial 9



# TUTORIAL 8

2. An asteroid approaches a planet from infinity, in a parabolic orbit, and barely touches the surface of the planet at its closest approach, before moving away again. If the mass density (mass per unit volume) of the planet is $\rho_m$ (assumed uniform throughout), determine the angular velocity of the asteroid as it passes the planet.

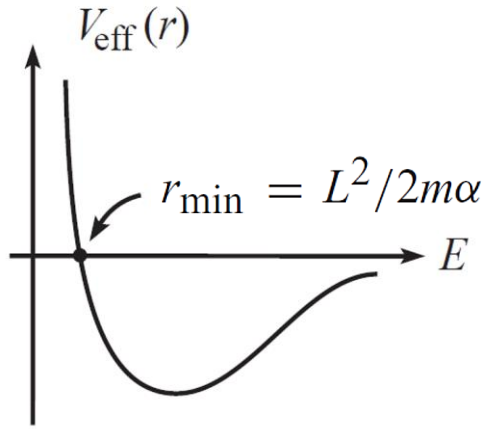



- **Parabola** ($\epsilon = 1$)

$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon\cos\theta),$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}}$$

$$\alpha \equiv GM_{\odot}m$$



# TUTORIAL 8

6. Find the force law for a central force field that allows a particle to move in a logarithmic orbit $r = ke^{\alpha\theta}$, where $k$ and $\alpha$ are constants. Then find $\theta(t)$ and $r(t)$.

$$\left(\frac{1}{r^2}\frac{dr}{d\theta}\right)^2 = \frac{2mE}{L^2} - \frac{1}{r^2} - \frac{2mV(r)}{L^2}$$

$$\mathbf{F}(\mathbf{r}) = -\nabla V(r) = -\frac{dV}{dr}\hat{\mathbf{r}}$$

# VECTOR CALCULUS

- **∇** Operator: A vector operator

$$\nabla = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}.$$

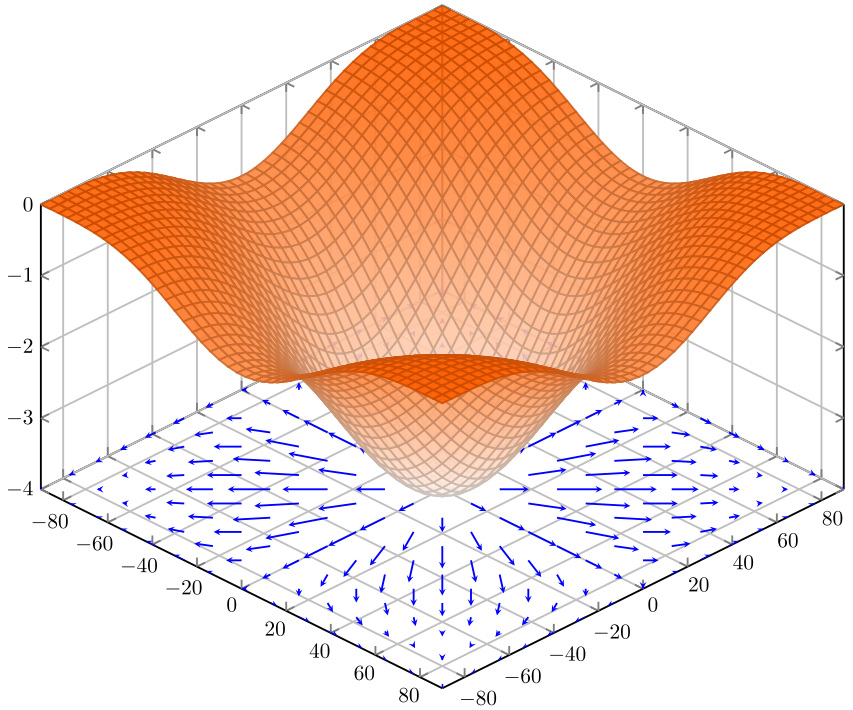
- When acts on a scalar – $\nabla T$: **Gradient**
- When acts on a vector via the dot product – $\nabla . T$: **Divergence**
- When acts on a vector via the cross product– $\nabla \times T$: **Curl**

# VECTOR CALCULUS

- Gradient

$$\nabla T \equiv \frac{\partial T}{\partial x}\hat{\mathbf{x}} + \frac{\partial T}{\partial y}\hat{\mathbf{y}} + \frac{\partial T}{\partial z}\hat{\mathbf{z}}$$

- The magnitude $|\nabla T|$ gives the slope (rate of increase) along the maximal direction
- The gradient $\nabla T$ points in the direction of maximum increase of the function T

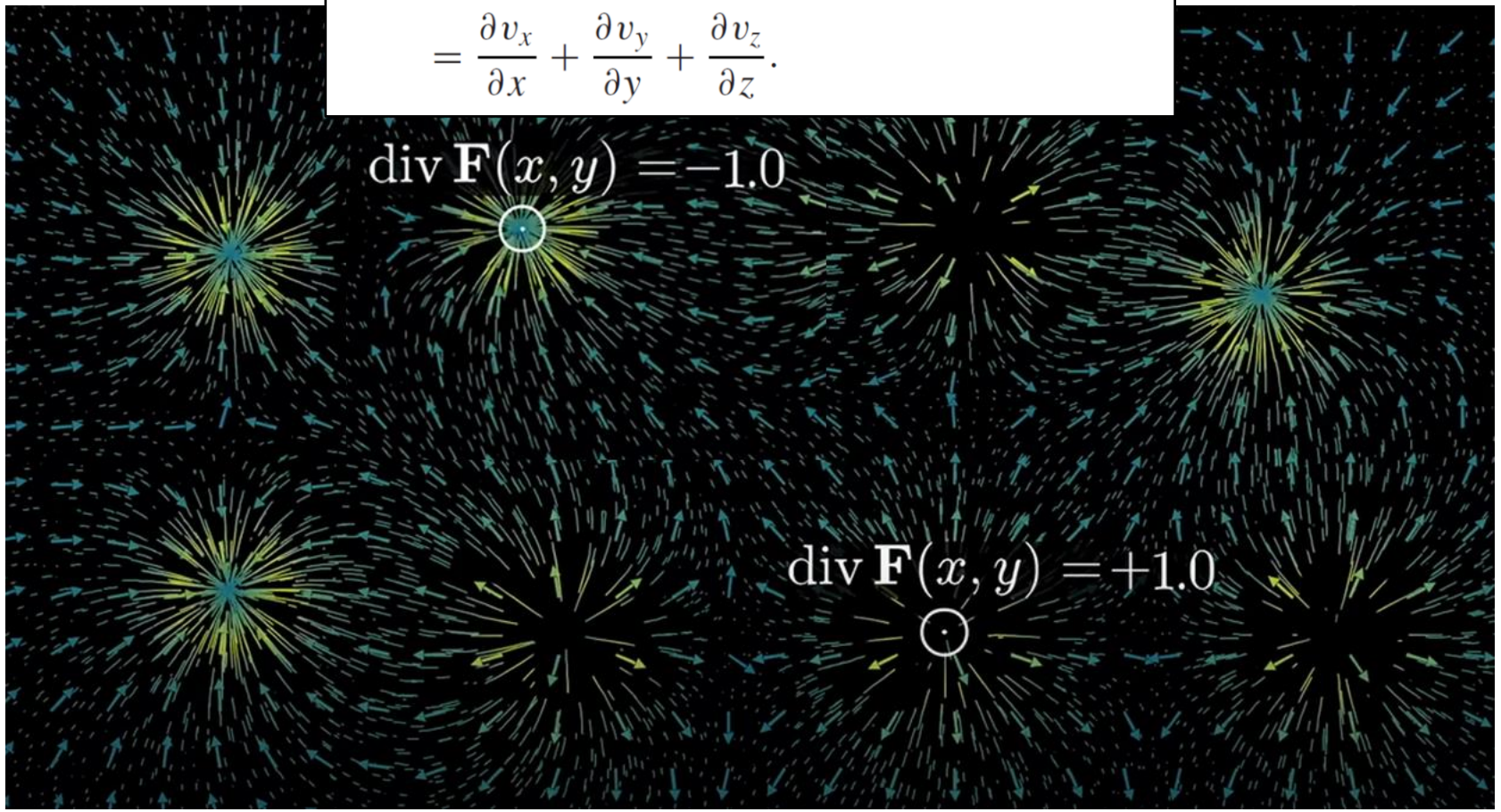



**A 2D function**

# VECTOR CALCULUS

- Divergence

$$\nabla \cdot \mathbf{v} = \left(\hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}\right) \cdot (v_x\hat{\mathbf{x}} + v_y\hat{\mathbf{y}} + v_z\hat{\mathbf{z}})$$
$$= \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}.$$



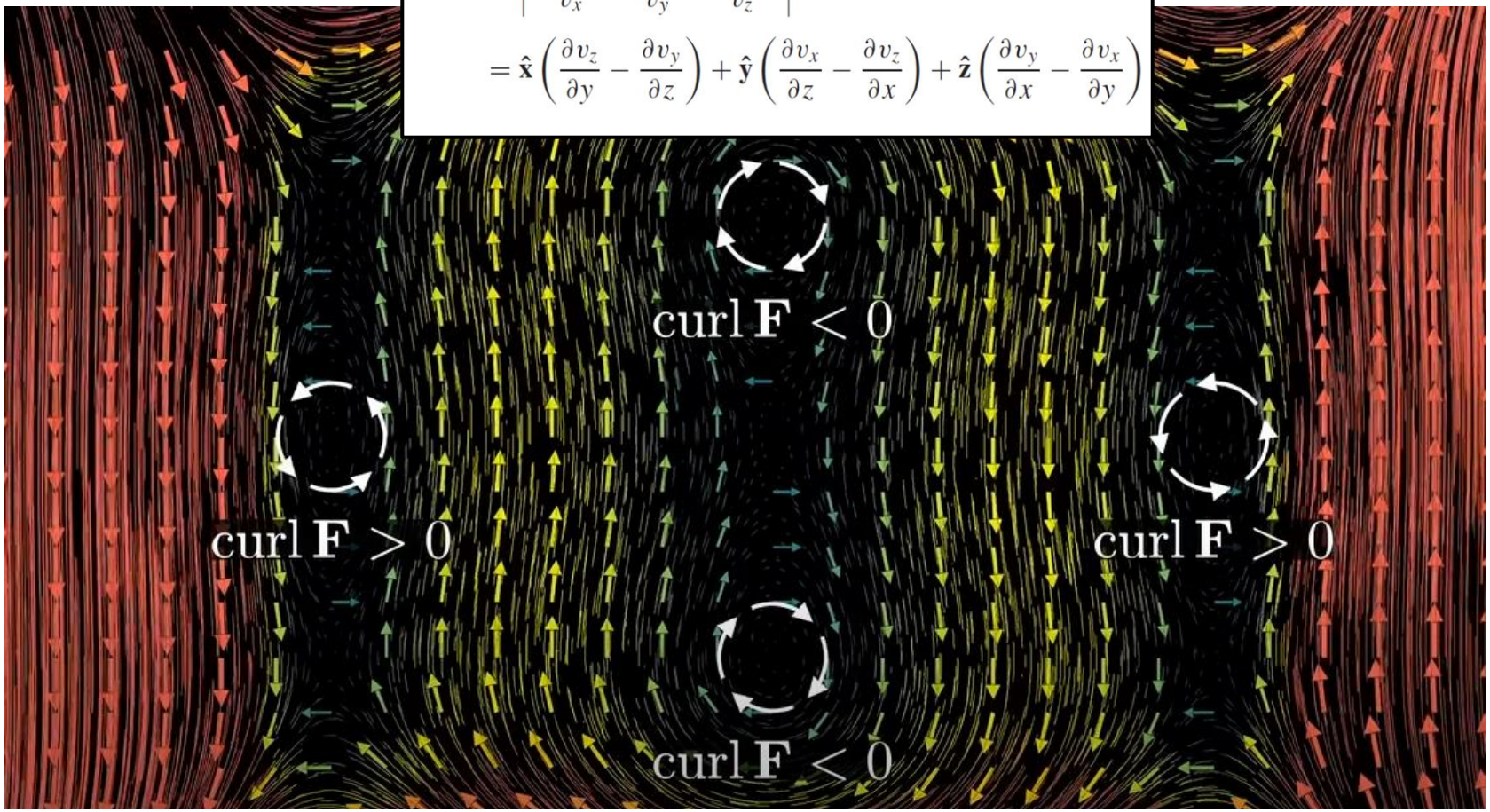

How much the vector spreads out – a scalar quantity

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# VECTOR CALCULUS

- Curl

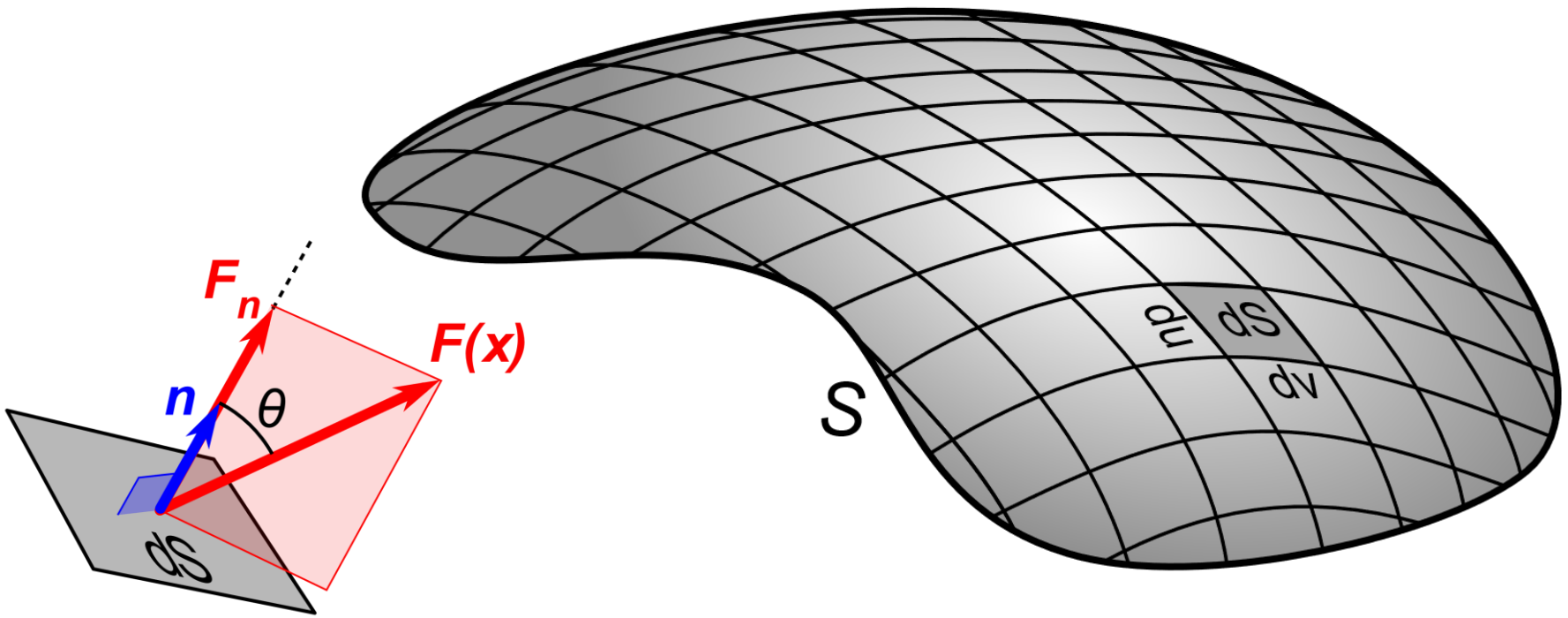
$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix}$$
$$= \hat{\mathbf{x}}\left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)$$




How much the vector curls around – a vector quantity

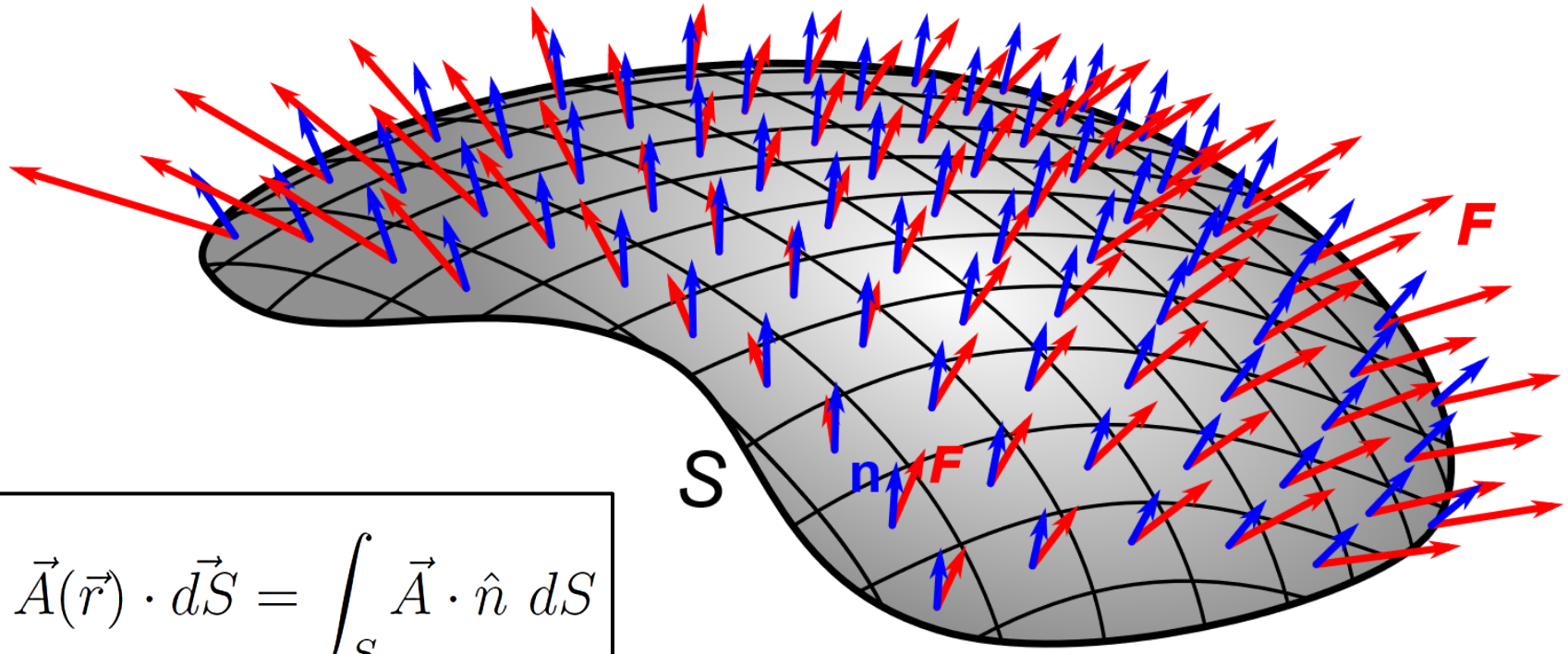
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# VECTOR FIELD: SURFACE INTEGRALS




The flux through each patch is equal to the normal (perpendicular) component of the field $F_n(x) = F(x)cos\theta$ at the patch's location $x$ multiplied by the area $dS$.

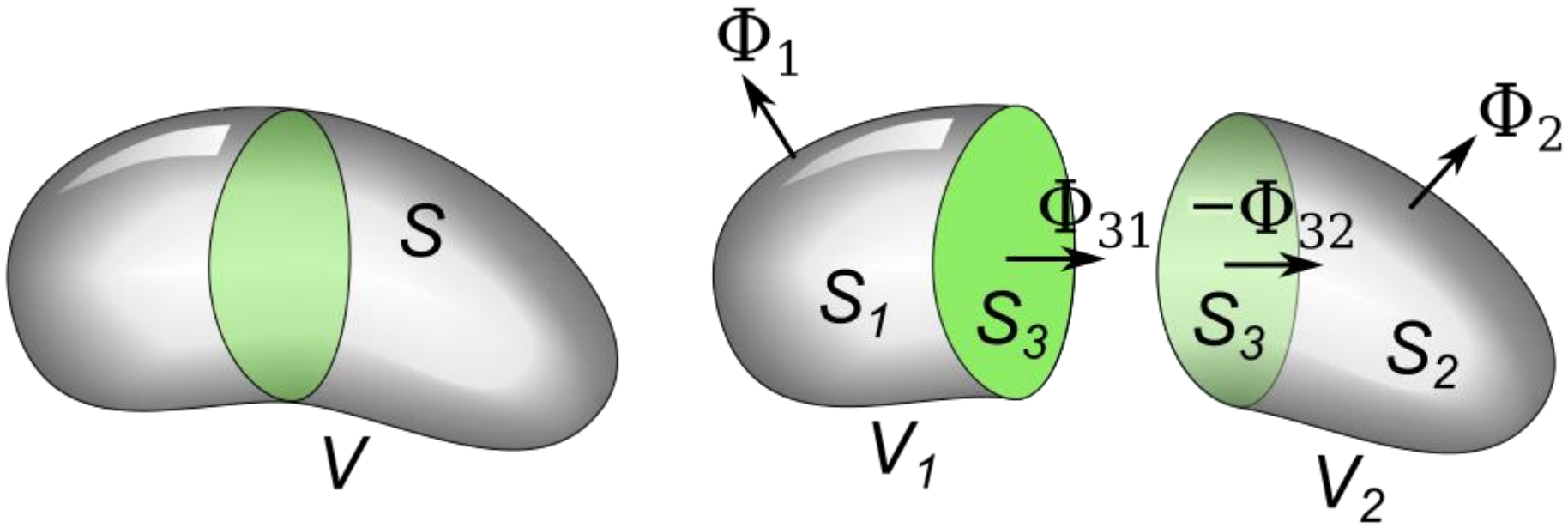
# VECTOR FIELD: SURFACE INTEGRALS




$$\Phi = \int_S \vec{A}(\vec{r}) \cdot d\vec{S} = \int_S \vec{A} \cdot \hat{n}\, dS$$

The total flux through the surface is found by adding up $\boldsymbol{F} \cdot \hat{\boldsymbol{n}}\, dS$ for each patch. In the limit as the patches become infinitesimally small, this is the surface integral!

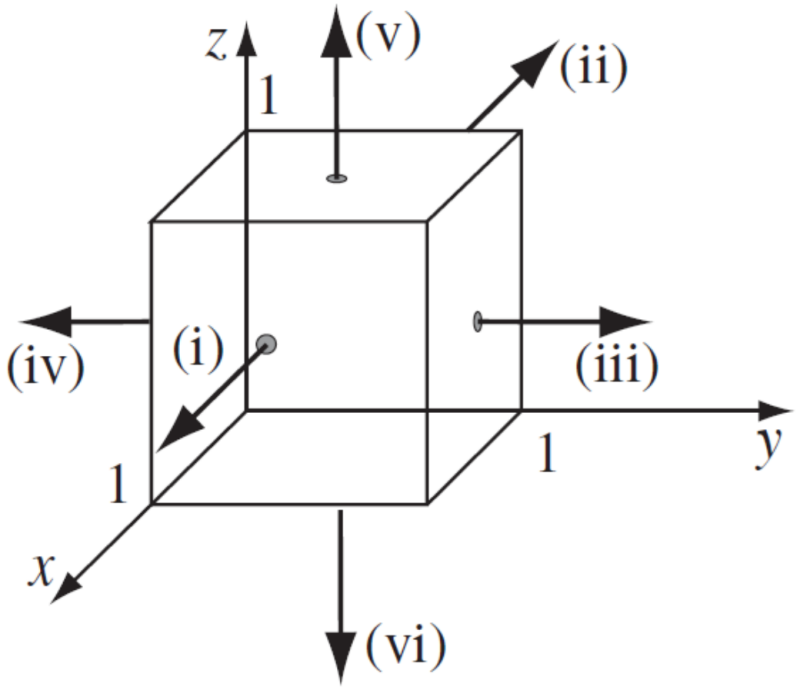
# VECTOR FIELD: SURFACE INTEGRALS




$$\int_S \vec{A} \cdot d\vec{S} = \int_{S_1} \vec{A} \cdot d\vec{S} + \int_{S_2} \vec{A} \cdot d\vec{S}$$

# VECTOR FIELD: DIVERGENCE THEOREM

It enables us to write the surface integral of any vector field over a closed surface S as the volume integral of the divergence of the field over the volume of space enclosed by S.



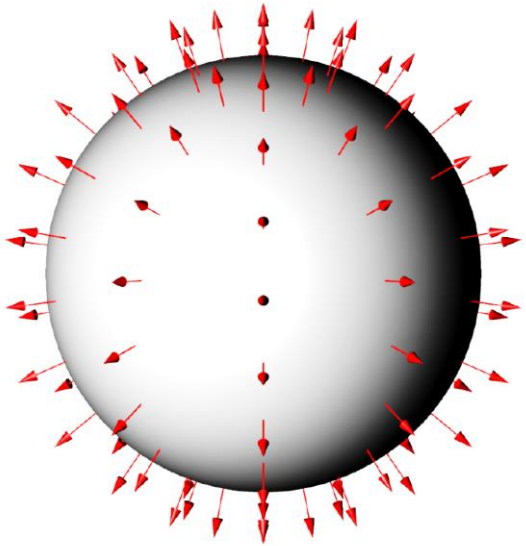

How much the vector spreads out

Flux

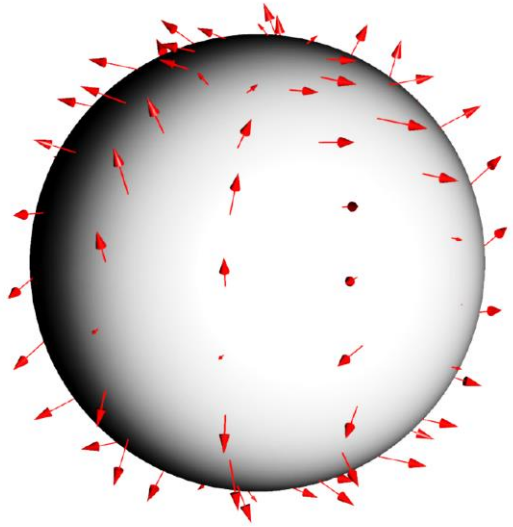
$$\int_{\mathcal{V}} (\nabla \cdot \mathbf{v})\, d\tau = \oint_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a}.$$

# VECTOR FIELD: DIVERGENCE THEOREM

$$\mathbf{F}(x,y,z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$



$$\mathbf{F}(x,y,z) = xz^2\mathbf{i} + yx^2\mathbf{j} + zy^2\mathbf{k}$$

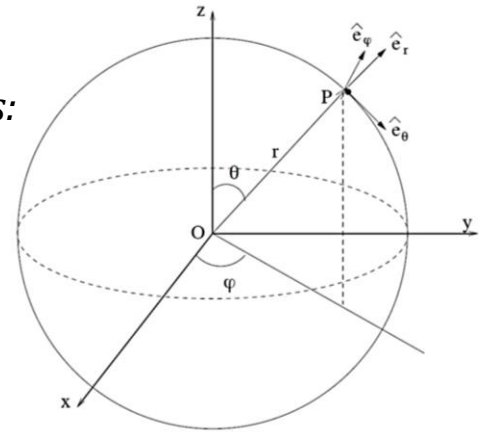


# VECTOR FIELD: GRADIENT (RECAP)

## Spherical Polar

*Relations with the cartesian coordinates:*

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$




*Unit vectors:*

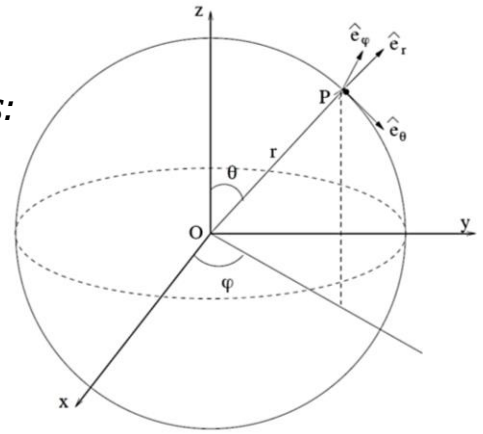
$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

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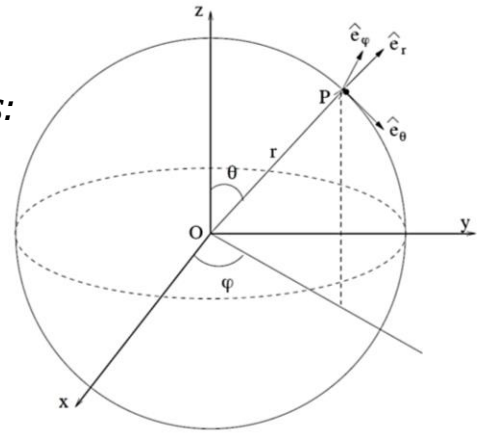
$$\hat{\mathbf{x}} = \sin\theta\cos\varphi\hat{\mathbf{r}} + \cos\theta\cos\varphi\hat{\boldsymbol{\theta}} - \sin\varphi\hat{\boldsymbol{\varphi}}$$
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# VECTOR FIELD: GRADIENT (RECAP)

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$$\vec{\nabla} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}$$

*Unit vectors:*

$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
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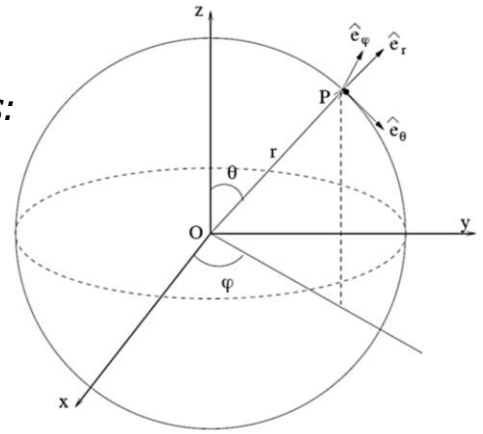
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# VECTOR FIELD: GRADIENT (RECAP)

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$$\vec{\nabla} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} = a\hat{r}\frac{\partial}{\partial r} + b\hat{\theta}\frac{\partial}{\partial\theta} + c\hat{\phi}\frac{\partial}{\partial\phi}$$

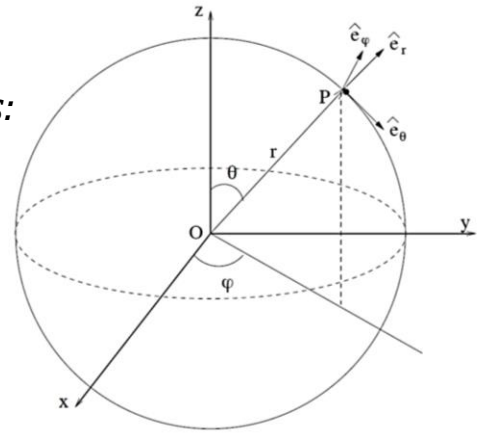
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# VECTOR FIELD: GRADIENT (RECAP)

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$$\hat{\mathbf{x}} = \sin\theta\cos\varphi\hat{\mathbf{r}} + \cos\theta\cos\varphi\hat{\boldsymbol{\theta}} - \sin\varphi\hat{\boldsymbol{\varphi}}$$
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$$\vec{\nabla} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} = a\hat{r}\frac{\partial}{\partial r} + b\hat{\theta}\frac{\partial}{\partial\theta} + c\hat{\phi}\frac{\partial}{\partial\phi}$$

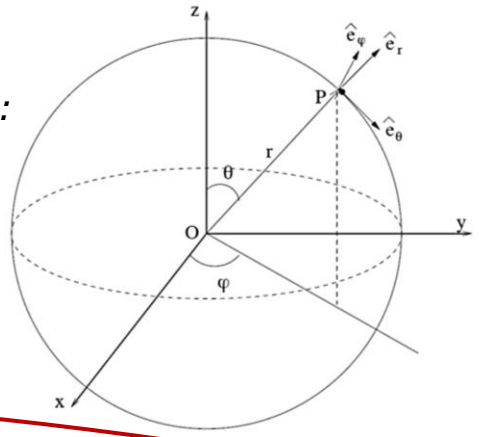
$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r}\frac{\partial}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial}{\partial y} + \frac{\partial z}{\partial r}\frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial\theta} = \frac{\partial x}{\partial\theta}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\theta}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\theta}\frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\phi}\frac{\partial}{\partial z}$$

# VECTOR FIELD: GRADIENT (RECAP)

## Spherical Polar

*Relations with the cartesian coordinates:*

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$$\hat{\mathbf{x}} = \sin\theta\cos\varphi\hat{\mathbf{r}} + \cos\theta\cos\varphi\hat{\boldsymbol{\theta}} - \sin\varphi\hat{\boldsymbol{\varphi}}$$
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$$\vec{\nabla} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} = a\hat{r}\frac{\partial}{\partial r} + b\hat{\theta}\frac{\partial}{\partial\theta} + c\hat{\phi}\frac{\partial}{\partial\phi}$$

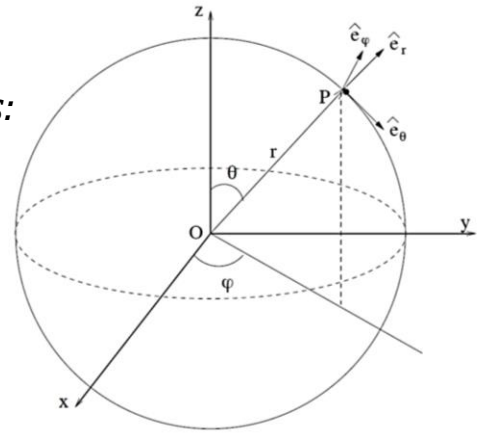
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$$\frac{\partial}{\partial\theta} = \frac{\partial x}{\partial\theta}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\theta}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\theta}\frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\phi}\frac{\partial}{\partial z}$$

# VECTOR FIELD: GRADIENT (RECAP)

## Spherical Polar

*Relations with the cartesian coordinates:*

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*Unit vectors:*

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$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
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$$\hat{\mathbf{x}} = \sin\theta\cos\varphi\hat{\mathbf{r}} + \cos\theta\cos\varphi\hat{\boldsymbol{\theta}} - \sin\varphi\hat{\boldsymbol{\varphi}}$$
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$$\vec{\nabla} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} = a\hat{r}\frac{\partial}{\partial r} + b\hat{\theta}\frac{\partial}{\partial\theta} + c\hat{\phi}\frac{\partial}{\partial\phi}$$

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r}\frac{\partial}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial}{\partial y} + \frac{\partial z}{\partial r}\frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial\theta} = \frac{\partial x}{\partial\theta}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\theta}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\theta}\frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\phi}\frac{\partial}{\partial z}$$

$$\boldsymbol{\nabla} f = \frac{\partial f}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial f}{\partial\theta}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial\varphi}\hat{\boldsymbol{\varphi}}$$

# VECTOR FIELD: DIVERGENCE

## Spherical Polar

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \arctan\left(\frac{\sqrt{x^2 + y^2}}{z}\right)$$

$$\varphi = \arctan\left(\frac{y}{x}\right)$$

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial x} + \frac{\partial}{\partial \phi}\frac{\partial \phi}{\partial x}$$

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial y} + \frac{\partial}{\partial \phi}\frac{\partial \phi}{\partial y}$$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial z} + \frac{\partial}{\partial \phi}\frac{\partial \phi}{\partial z}$$

$$\nabla \cdot \vec{A} = \frac{\partial}{\partial x}(A_x) + \frac{\partial}{\partial y}(A_y) + \frac{\partial}{\partial z}(A_z)$$

$$A_x = \sin\theta\cos\phi A_r + \cos\theta\cos\phi A_\theta - \sin\phi A_\phi$$

$$A_y = sin\theta sin\phi A_r + cos\theta sin\phi A_\theta + cos\phi A_\phi$$

$$A_z = \cos\theta A_r - \sin\theta A_\theta$$

$$\boldsymbol{\nabla \cdot A} = \frac{1}{r^2}\frac{\partial\left(r^2 A_r\right)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(A_\theta \sin\theta\right) + \frac{1}{r\sin\theta}\frac{\partial A_\varphi}{\partial\varphi}$$

**Area element:** $\vec{dS} = (R^2 \sin\theta\ d\theta\ d\varphi)\ \hat{e}_r$

**Volume element:** $dV = r^2 \sin\theta\ dr\ d\theta\ d\varphi$

# VECTOR FIELD: DIVERGENCE

## Cylindrical

$$\rho = \sqrt{x^2 + y^2}$$

$$\varphi = \arctan\left(\frac{y}{x}\right)$$

$$z = z$$

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \rho}\frac{\partial \rho}{\partial x} + \frac{\partial}{\partial \emptyset}\frac{\partial \emptyset}{\partial x}$$

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial \rho}\frac{\partial \rho}{\partial y} + \frac{\partial}{\partial \emptyset}\frac{\partial \emptyset}{\partial y}$$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial z}$$

$$\nabla \cdot \vec{A} = \frac{\partial}{\partial x}(A_x) + \frac{\partial}{\partial y}(A_y) + \frac{\partial}{\partial z}(A_z)$$

$$A_x = \cos \emptyset \, A_\rho - \sin \emptyset \, A_\emptyset$$

$$A_y = \sin \emptyset \, A_\rho + \cos \emptyset \, A_\emptyset$$

$$A_z = A_z$$

$$\boldsymbol{\nabla \cdot A} = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

**Area element:** $\vec{dS} = \rho \, d\rho \, d\varphi \, \hat{e}_z, \rho \, d\varphi \, dz \, \hat{e}_\rho \text{ and } d\rho \, dz \, \hat{e}_\phi$

**Volume element:** $dV = (d\rho)\,(\rho d\varphi)\,(dz) = \rho \, d\rho \, d\varphi \, dz$

# VECTOR FIELD: CURVILINEAR COORDINATES

## Gradient, divergence and curl in a generic set of orthogonal coordinates $(q_1, q_2, q_3)$

**Cartesian coordinates**

$x_1 = x, x_2 = y, x_3 = z$

$h_1 = 1, h_2 = 1, h_3 = 1,$

**Cylindrical coordinates**

$x_1 = r, x_2 = \phi, x_3 = z$

$h_1 = 1, h_2 = r, h_3 = 1$

**Spherical coordinates**

$x_1 = r, x_2 = \theta, x_3 = \phi$

$h_1 = 1, h_2 = r, h_3 = r\sin\theta$

- Consider the line element: $d\vec{r} = \sum_i^3 \frac{\partial \vec{r}}{\partial q_i} dq_i = \sum_{i=1}^3 \left|\frac{\partial \vec{r}}{\partial q_i}\right| dq_i \hat{q}_i = \sum_{i=1}^3 h_i dq_i \hat{q}_i$
- Surface area $dS = h_1 h_2 \delta q_1 \delta q_2$ etc. Volume element: $dV = h_1 h_2 h_3 \delta q_1 \delta q_2 \delta q_3$

$$\vec{\nabla} f = \frac{1}{h_1}\frac{\partial f}{\partial q_1} + \frac{1}{h_2}\frac{\partial f}{\partial q_2} + \frac{1}{h_3}\frac{\partial f}{\partial q_3}$$

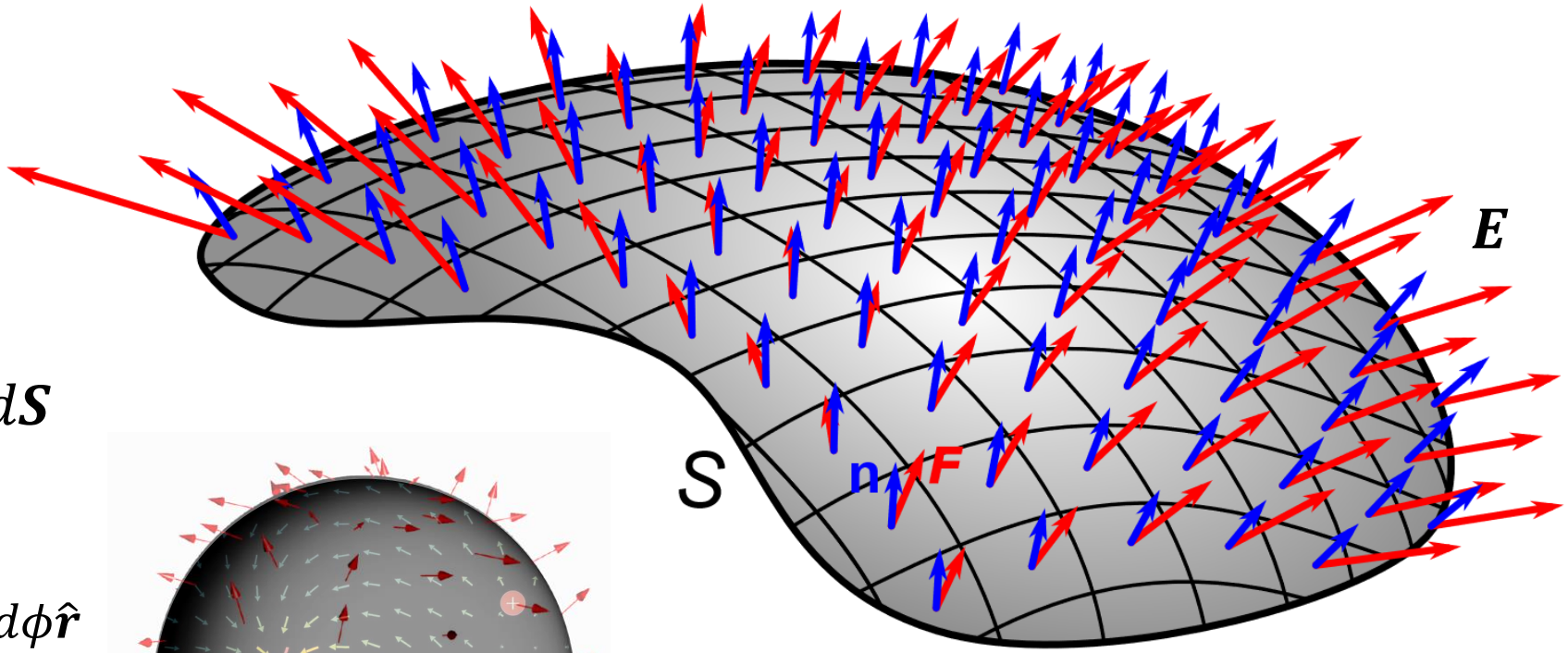
$$\vec{\nabla} \cdot \vec{A} = \frac{1}{h_1 h_2 h_3}\left[\partial_{q_1}(h_2 h_3 A_1) + \partial_{q_2}(h_1 h_3 A_2) + \partial_{q_3}(h_1 h_2 A_3)\right]$$

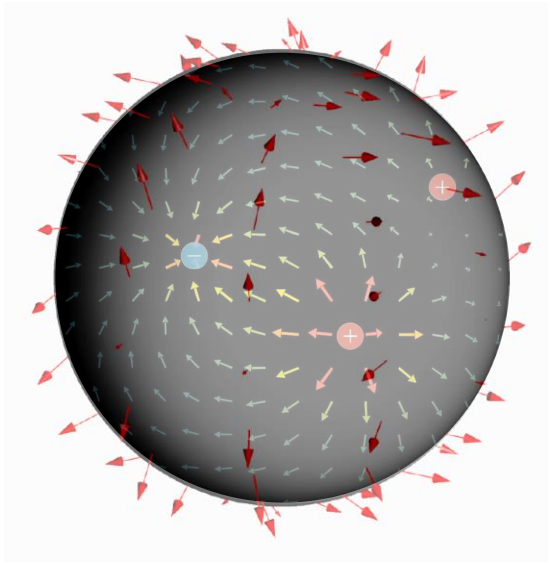
$$\vec{\nabla} \times \vec{A} = \frac{1}{h_1 h_2 h_3}\begin{vmatrix} \hat{q}_1 h_1 & \hat{q}_2 h_2 & \hat{q}_3 h_3 \\ \partial_{q_1} & \partial_{q_2} & \partial_{q_3} \\ A_1 h_1 & A_2 h_2 & A_3 h_3 \end{vmatrix}$$

# VECTOR FIELD: ELECTRIC FLUX

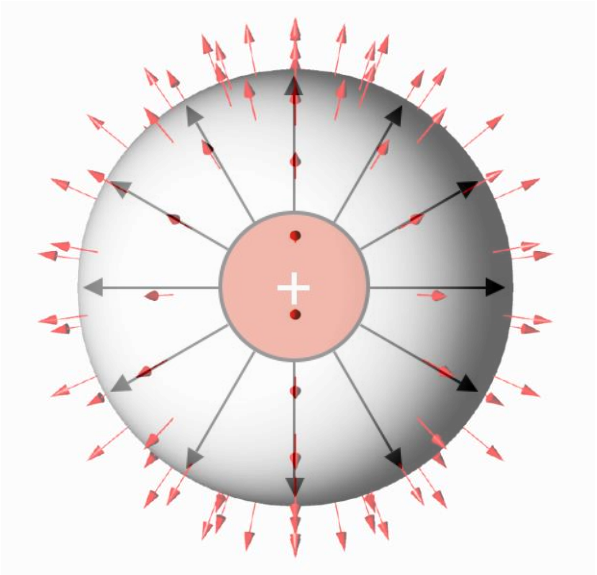
$$\Phi_E = \oint_S \boldsymbol{E} \cdot d\boldsymbol{S}$$

$$d\boldsymbol{S} = R^2 \sin\theta \, d\theta d\phi \hat{\boldsymbol{r}}$$

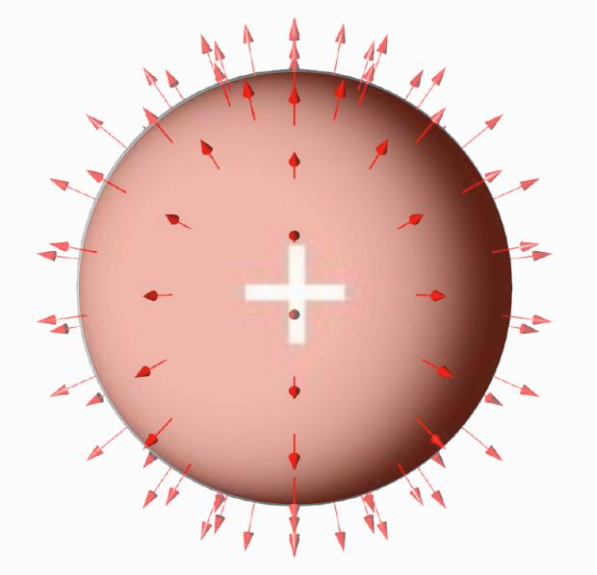


# VECTOR FIELD: GAUSS'S LAW



$$\oint \boldsymbol{E}.d\boldsymbol{S} = \frac{1}{\epsilon_0} q$$



$$\oint \boldsymbol{E}.d\boldsymbol{S} = \frac{1}{\epsilon_0} Q_{enc}$$

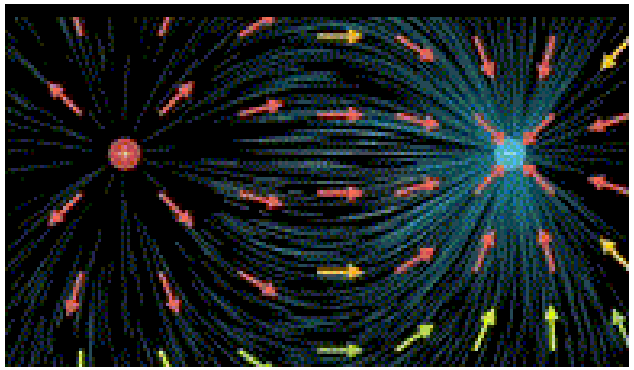
# VECTOR FIELD: GAUSS'S LAW

How much the vector spreads out

Flux

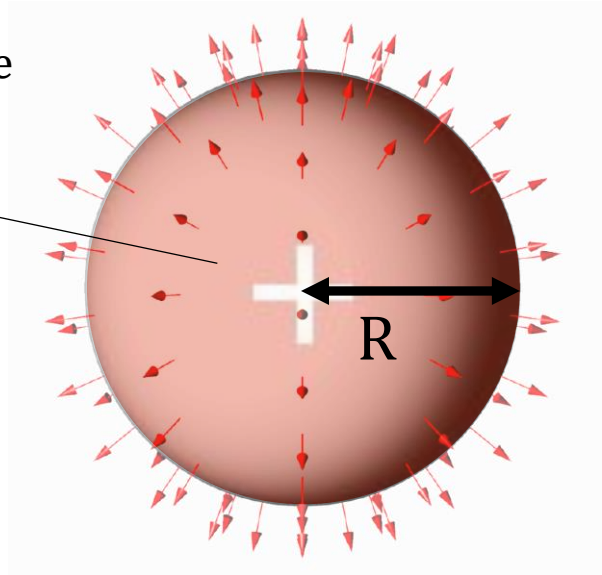
$$\int_{\mathcal{V}} (\nabla \cdot \mathbf{v})\, d\tau = \oint_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a}.$$

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}\rho.$$



Uniform charge density

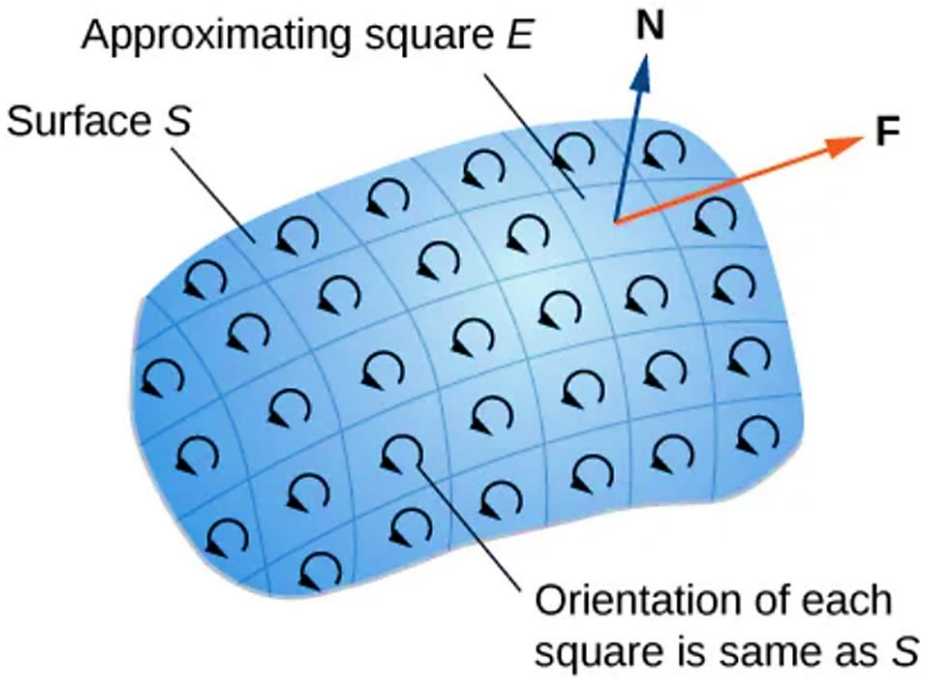
$\rho$

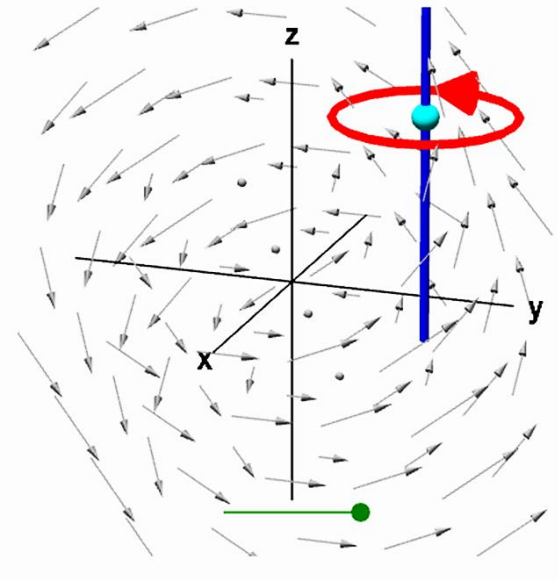



$$\oint \boldsymbol{E}.d\boldsymbol{S} = \frac{1}{\epsilon_0} Q_{enc}$$

Incompressible fluid flow with a source and a sink!

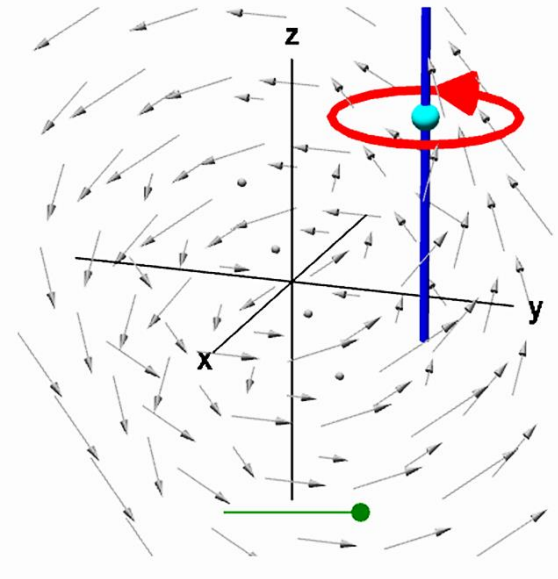
# VECTOR FIELD: STOKES' THEOREM



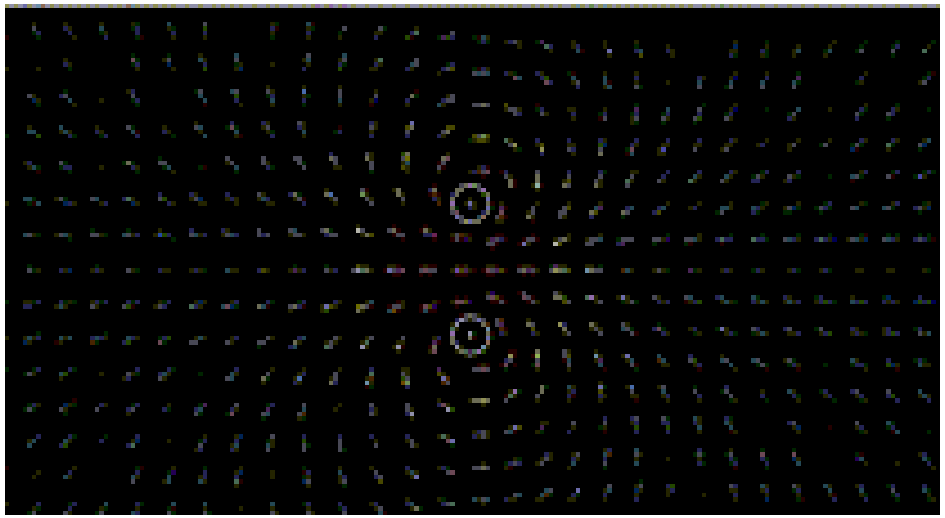
# VECTOR FIELD: STOKES' THEOREM

$$\oint_C \vec{A} \cdot \vec{dl} = \int_S \text{curl } \vec{A} \cdot \vec{dS}$$



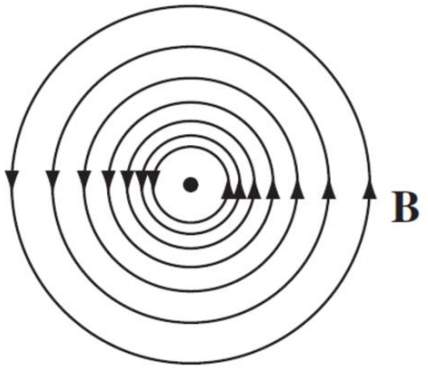

It enables us to rewrite the circulation of a vector field over a closed curve C as the surface integral of curl of the field over a surface S that is bounded by C.

# FLUID DYNAMICS: MAGNETOSTATICS



$$\nabla \cdot \mathbf{B} = 0.$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J},$$



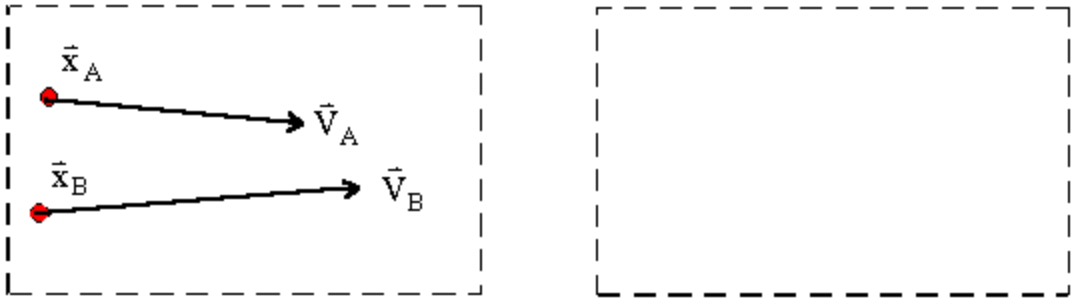

Magnetostatic field of a long wire

- Incompressible fluid flow without a source and a sink – Magnetic monopoles do not exist!
- Magnetic field lines always come in complete loops, rather than starting or ending at a point.

# FLUID DYNAMICS: INVISCID FLUID

Incompressible , homogeneous (no sources or sinks) fluid flow with zero viscosity (no dissipative force)

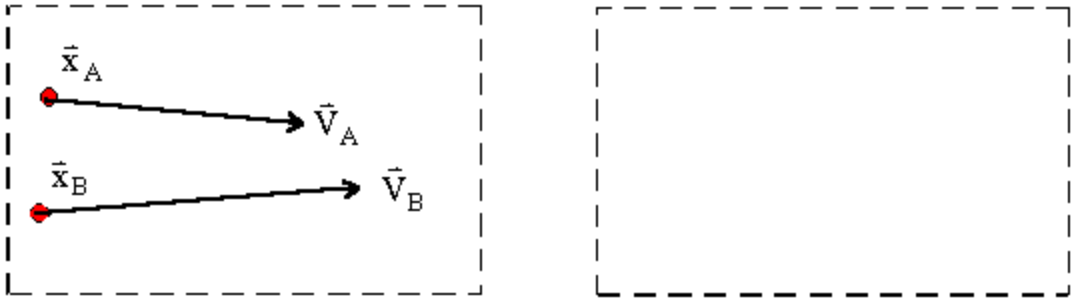
**Material Description**



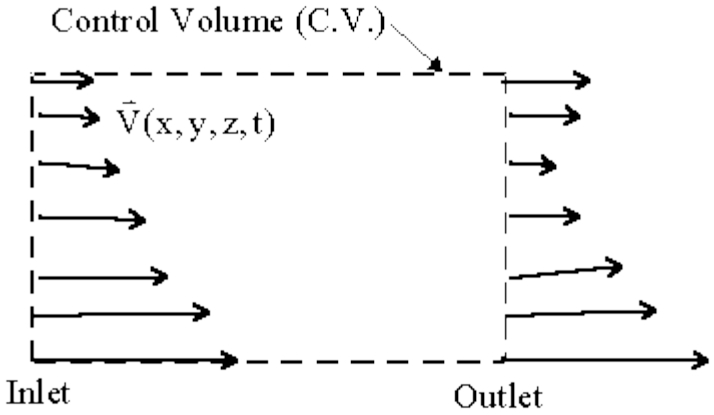
# FLUID DYNAMICS: INVISCID FLUID

Incompressible , homogeneous (no sources or sinks) fluid flow with zero viscosity (no dissipative force)

**Material Description**




**Spatial Description**



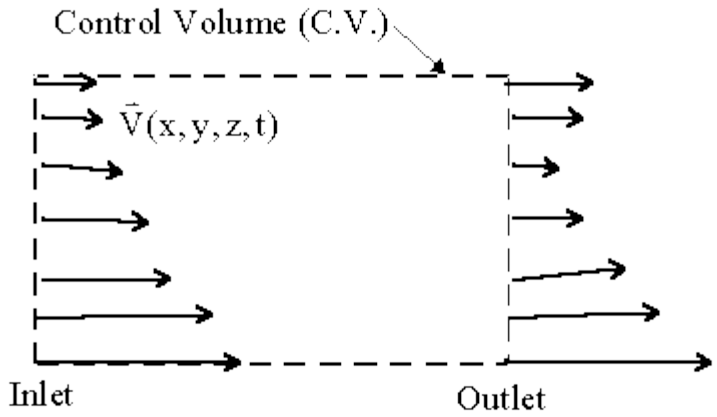
# FLUID DYNAMICS: INVISCID FLUID

Incompressible , homogeneous (no sources or sinks) fluid flow with zero viscosity (no dissipative force)

Velocity field $\boldsymbol{v}(\boldsymbol{r}, t)$ is fluids velocity at a point $\boldsymbol{r}$, acceleration we want is the rate of change of $\boldsymbol{v}(\boldsymbol{r}, t)$ of a given piece of fluid as it moves!

$$\frac{d\varrho}{dt} = \frac{\partial \varrho}{\partial t} + \mathbf{v} \cdot \nabla \varrho$$

**Spatial Description**



# FLUID DYNAMICS: INVISCID FLUID

Incompressible , homogeneous (no sources or sinks) fluid flow with zero viscosity (no dissipative force)

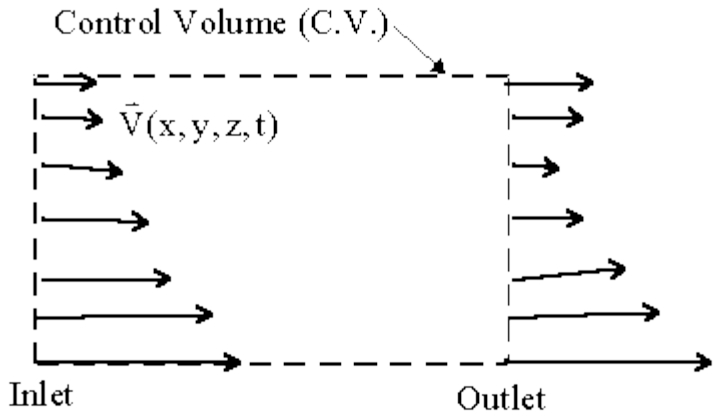
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$$\frac{d\mathbf{v}}{dt} = \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v}.$$

All three components put together!

**Spatial Description**



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$$\frac{d\mathbf{v}}{dt} = \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v}.$$

All three components put together!

**Forces present**:

1. Pressure gradient force arising from the difference in pressure across the element.
2. Some conservative force (due to gravity)

$$(\rho dV)\frac{d\boldsymbol{v}}{dt} = \boldsymbol{F}_{total} = \boldsymbol{F}_{gravity} + \boldsymbol{F}_{pressure}$$

# FLUID DYNAMICS: INVISCID FLUID

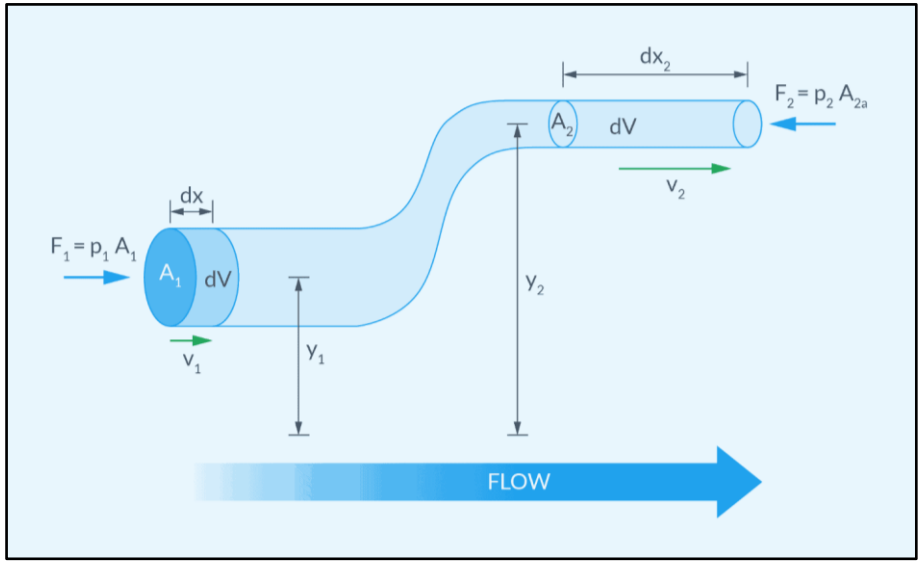
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All three components put together!




$$(\rho dV)\frac{d\boldsymbol{v}}{dt} = \boldsymbol{F}_{total} = \boldsymbol{F}_{gravity} + \boldsymbol{F}_{pressure}$$

# FLUID DYNAMICS: INVISCID FLUID

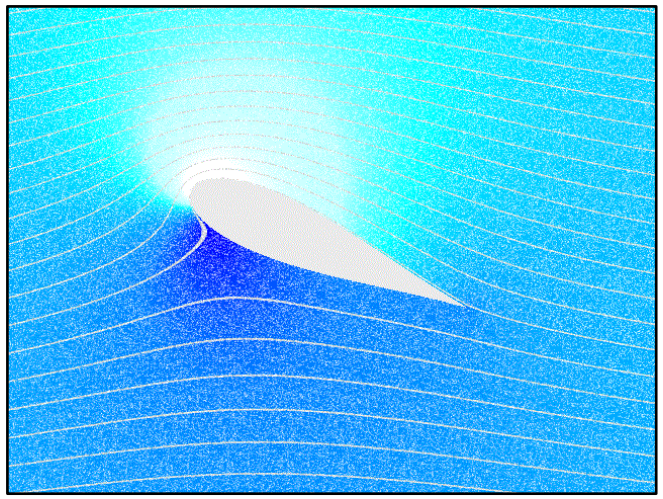
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All three components put together!



Equation of motion

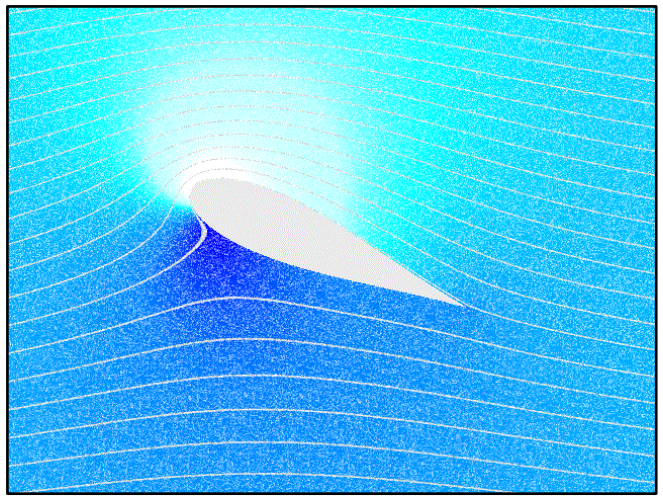
$$\rho \frac{d\boldsymbol{v}}{dt} = \rho \boldsymbol{g} - \boldsymbol{\nabla} p$$

# FLUID DYNAMICS: INVISCID FLUID

Incompressible , homogeneous (no sources or sinks) fluid flow with zero viscosity (no dissipative force)

$$\frac{1}{2}\varrho\frac{dv^2}{dt}+\varrho\frac{d(gz)}{dt}+\frac{dp}{dt}=0.$$

$$\frac{d}{dt}\left(\tfrac{1}{2}\varrho v^2+\varrho gz+p\right)=0.$$



Equation of motion

$$\rho\frac{d\boldsymbol{v}}{dt}=\rho\boldsymbol{g}-\boldsymbol{\nabla}p$$

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Incompressible , homogeneous (no sources or sinks) fluid flow with zero viscosity (no dissipative force)

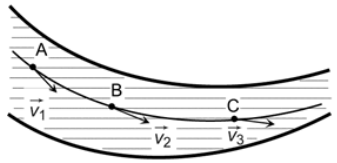
*Streamlines are a family of curves whose tangent vectors constitute the velocity vector field of the flow.*

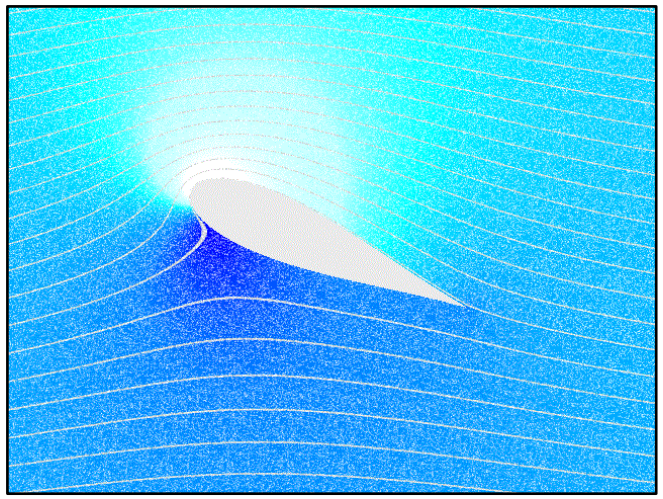
**Bernoulli's principle**

$$\frac{1}{2}\varrho\frac{dv^2}{dt}+\varrho\frac{d(gz)}{dt}+\frac{dp}{dt}=0.$$

$$\frac{d}{dt}\left(\tfrac{1}{2}\varrho v^2+\varrho gz+p\right)=0.$$

A constant along any streamline!

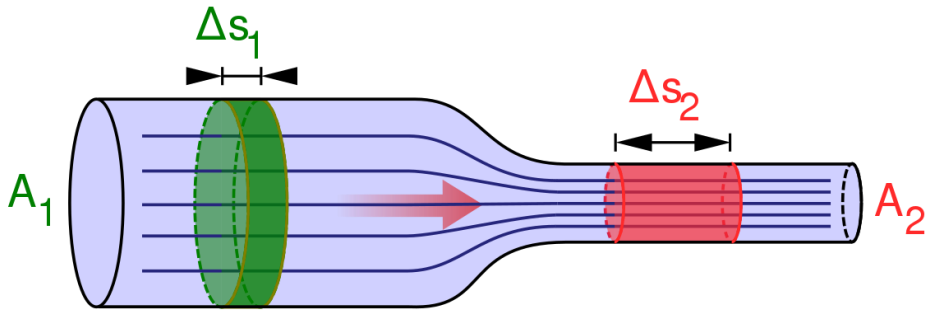


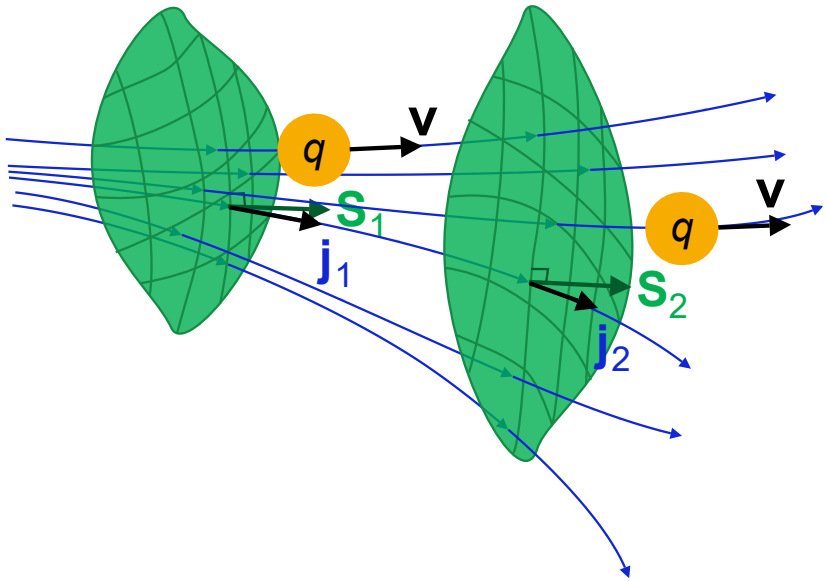



Equation of motion

$$\rho\frac{d\boldsymbol{v}}{dt}=\rho\boldsymbol{g}-\boldsymbol{\nabla}p$$

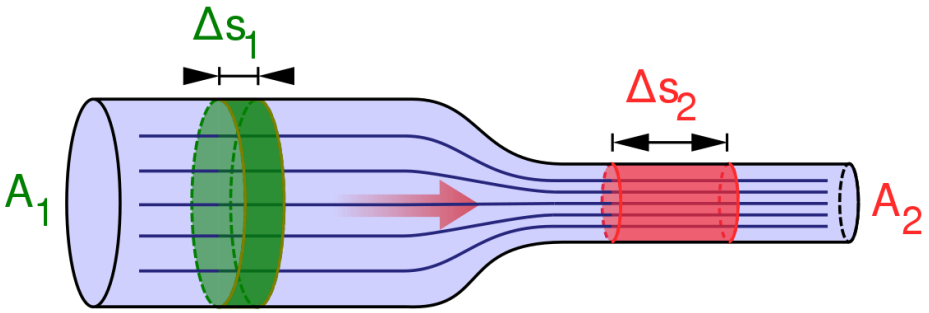
# FLUID DYNAMICS: CONTINUITY EQUATION

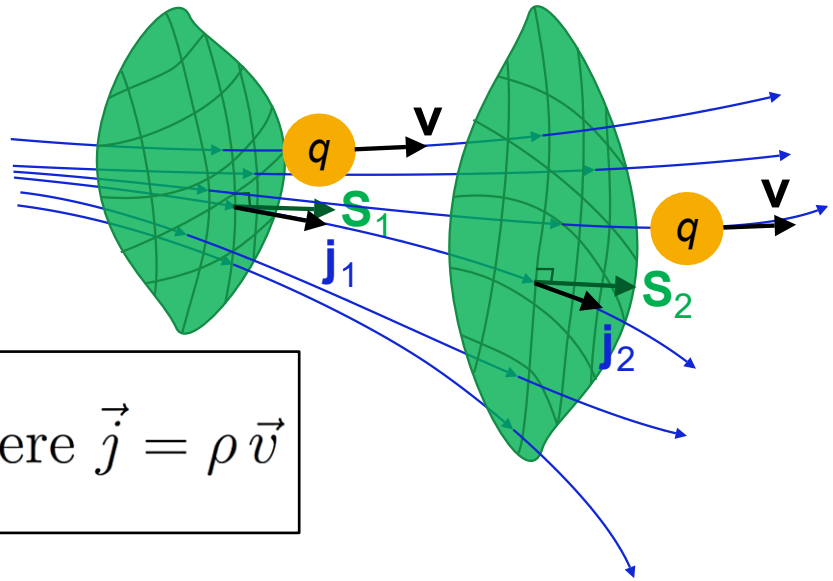



- Consider the flow of a fluid or of electric charge.
- $\rho(\vec{r},t)$ is charge density (or mass density of the fluid).
- $\vec{J}(\vec{r},t)$ is the corresponding current density crossing unit area per unit time.
- The flux of $\vec{J}$ over a closed surface equals the rate at which charge (or mass) leaves the volume enclosed by surface.

# FLUID DYNAMICS: CONTINUITY EQUATION

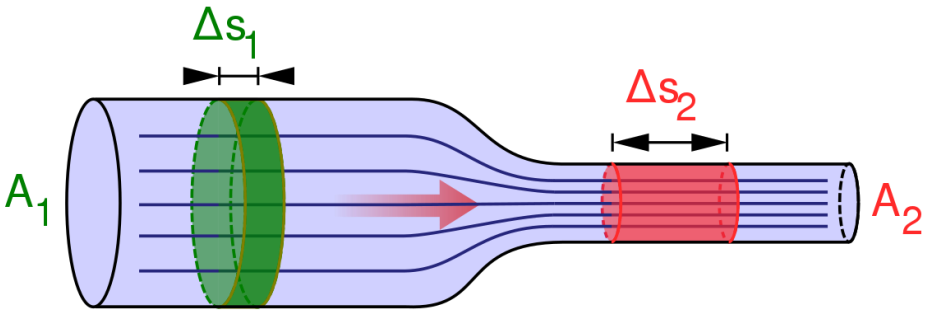


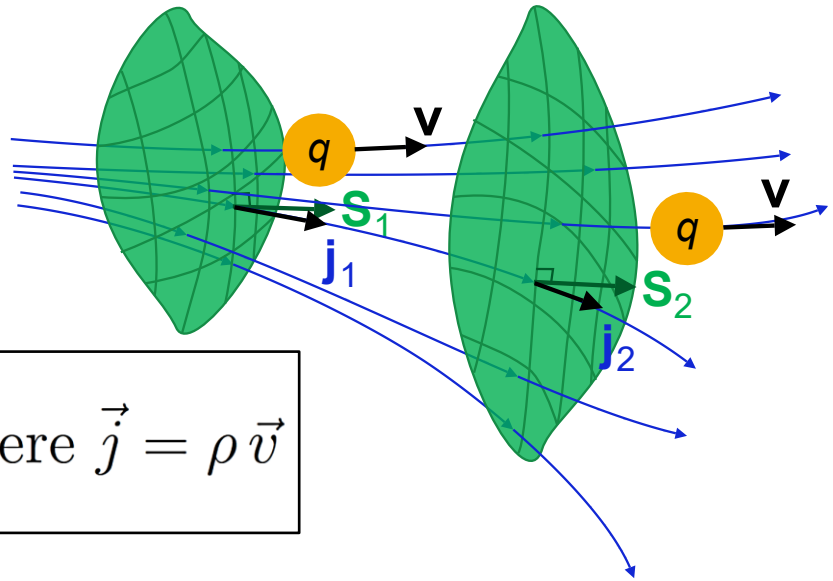




$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0\,, \quad \text{where } \vec{j} = \rho\,\vec{v}$$

Because charge is conserved, whatever flows out through the surface must come at the expense of what remains inside

# FLUID DYNAMICS: CONTINUITY EQUATION




$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0\,, \quad \text{where } \vec{j} = \rho\,\vec{v}$$

Because charge is conserved, whatever flows out through the surface must come at the expense of what remains inside

- The conservation law of a physical quantity is expressed as a continuity equation.
- It is “local” statement of conservation.

# TUTORIAL 9

1. **Gauss Theorem:** Check the divergence theorem for

   (a) the vector field

   $$\mathbf{V} = y^2\hat{\boldsymbol{x}} + (2xy + z^2)\hat{\boldsymbol{y}} + 2yz\hat{\boldsymbol{z}}$$

   and the unit cube situated at the origin.

   (b) the vector field

   $$\mathbf{V} = r\cos\theta\hat{\boldsymbol{r}} + r\sin\theta\hat{\boldsymbol{\theta}} + r\sin\theta\cos\varphi\hat{\boldsymbol{\varphi}}$$

   and the inverted hemispherical bowl of radius $R$ resting on the $xy$- plane and centered at the origin.

How much the vector spreads out

Flux

$$\int_{\mathcal{V}} (\nabla \cdot \mathbf{v})\, d\tau = \oint_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a}.$$

$$\begin{aligned}
\hat{e}_r &= \sin\theta\cos\varphi\ \hat{e}_x + \sin\theta\sin\varphi\ \hat{e}_y + \cos\theta\ \hat{e}_z \\
\hat{e}_\theta &= \cos\theta\cos\varphi\ \hat{e}_x + \cos\theta\sin\varphi\ \hat{e}_y - \sin\theta\ \hat{e}_z \\
\hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y \quad .
\end{aligned}$$

$$\boldsymbol{\nabla \cdot A} = \frac{1}{r^2}\frac{\partial\left(r^2 A_r\right)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(A_\theta \sin\theta\right) + \frac{1}{r\sin\theta}\frac{\partial A_\varphi}{\partial\varphi}$$

**Area element:** $\vec{dS} = (R^2 \sin\theta\ d\theta\ d\varphi)\ \hat{e}_r$

**Volume element:** $dV = r^2 \sin\theta\ dr\ d\theta\ d\varphi$

# TUTORIAL 9

2. **Stokes' Theorem:** The magnetic field due to a current density $\boldsymbol{J}$ is (Here $\mu_0$ is the permeability of the vacuum and $j_0$ and $a$ are constants of suitable dimension)

$$\boldsymbol{B}(\boldsymbol{r}) = \begin{cases} \frac{\mu_0 j_0}{2}(x\hat{\boldsymbol{y}} - y\hat{\boldsymbol{x}}) & \text{for } 0 \leq \rho \leq a, \\ \frac{\mu_0 j_0 a^2}{2\rho^2}(x\hat{\boldsymbol{y}} - y\hat{\boldsymbol{x}}) & \text{for } \rho \geq a. \end{cases}$$

$$x\hat{\boldsymbol{y}} - y\hat{\boldsymbol{x}} = \rho\hat{\boldsymbol{\varphi}}$$

Compute the circulation of the magnetic field over a circle of radius $R$ lying in the $xy$-plane centered at the origin, in two different ways:

$$\boldsymbol{dl} = Rd\varphi\hat{\boldsymbol{\varphi}}$$

(a) First, explicitly carry out the line integral.

(b) Next, evaluate the integral using Stokes' Theorem.

(c) Noting that $\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J}$, show that the circulation of $\boldsymbol{B}$ equals $\mu_0$ times the current through the planar surface enclosed by the circle. This is known as **Ampere's Law.**

$$\oint_C \vec{A} \cdot \vec{dl} = \int_S \text{curl } \vec{A} \cdot \vec{dS}$$

# TUTORIAL 9

3. **Continuity Equation:** The surface current density in the $xy$-plane is given by

$$\boldsymbol{K}(x,y) = K_0\left(\frac{x^2}{a^2}\hat{\boldsymbol{x}} + \frac{y^2}{b^2}\hat{\boldsymbol{y}}\right)$$

Evaluate the divergence of $\boldsymbol{K}$ and hence calculate the time rate of change of the charge contained in the first quadrant of a circle of unit radius centered on the origin. Show explicitly that for this quadrant, the continuity equation is satisfied in its integral form.

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0$$

$$\int_A (\boldsymbol{\nabla} \cdot \boldsymbol{K}) d^2r = \int_C \boldsymbol{K} \cdot \hat{\boldsymbol{n}} dl$$

# TUTORIAL 9

4. **Incompressible fluid:** A steady irrotational flow of an incompressible, nonviscous fluid in the gravitational field is described by a velocity potential given by $\phi(x, y, z) = kyz$ where k is positive constant. The positive z-direction is vertically upwards. The density of the fluid is $1000kg/m^2$.

   (a) Determine the velocity field.

   (b) Compute the pressure difference between two points $A(y = 1m, z = 1m)$ and $B(y = 2m, z = 2m)$.

   (c) Determine the equation of the streamlines.

$$\frac{d}{dt}\underbrace{\left(\tfrac{1}{2}\varrho v^2 + \varrho g z + p\right)}_{\text{A constant along any streamline!}} = 0.$$